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Forecasting fuel prices with the Chilean exchange rate: Going beyond the commodity currency hypothesis

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ABSTRACT

In this paper we show that the Chilean exchange rate has the ability to predict the returns of oil and of three additional oil-related products: gasoline, propane and heating oil. We show this using both in- and out-of sample exercises at multiple horizons. Natural explanations for our findings rely on the well know "dollar effect" and on the present-value theory for exchange rate determination in combination with the strong co-movement displayed by fuel and metal prices. Given that the Chilean economy is heavily influenced by copper, which represents nearly 50% of total national exports, the floating Chilean Peso is importantly affected by price fluctuations in this metal. As oil-related products display an important co-movement with base metal prices, it is reasonable to expect evidence of Granger causality from the Chilean peso to these oil-related products. Interestingly, we provide sound evidence indicating that the predictive ability of the Chilean Peso goes beyond these natural explanations. In particular, we show another plausible predictive channel: volatility in combination with a negative contemporaneous leverage effect in fuel returns. Finally, we compare the Chilean peso with other commodity-currencies in their ability to predict fuel returns. The Chilean peso fares extremely well in this competition, especially at short horizons of one, three and six months.

1. Introduction

In this paper we find that the Chilean exchange rate has a remarkable ability to forecast fuel returns. We show this by comparing population Mean Squared Prediction Errors (MSPE) between models with and without the Chilean Peso, using both in-sample and out-of sample exercises at multiple horizons of one, three, six, nine and twelve months ahead. Our results are interesting for three main reasons. First, we report that the currency of a net oil importer, that is not relevant in the international fuel markets, can predict fuel prices. Secondly, our analysis considers not just the frequently studied crude oil, but also, three oil-derivative products: propane, regular gasoline and heating oil. Third, we provide sound evidence indicating that the predictive ability of the Chilean Peso goes beyond the natural explanations based on a "dollar effect" or on the Commodity Currency Hypothesis (CCH). In particular, we show another plausible predictive channel: volatility in combination with a negative contemporaneous leverage effect in fuel returns. In addition, our models are effective and simple to implement, which is

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helpful for those that periodically need to update their fuel prices forecasts.

As mentioned before, two natural explanations of our findings rely on the well know "dollar effect" and on the present-value theory for exchange rate determination in combination with the strong co-movement displayed by fuel and commodity prices.[1] The first effect is related to the fact that fuel prices are denominated in US dollars. This implies that a higher value of this currency leads to a decrease in fuels' demand and a fall in their prices. The present-value theory for exchange rate determination implies that floating currencies from commodity exporters may Granger-cause future movements in international prices of primary commodity products. This implication is known as the CCH and has been under scrutiny since the birth of the papers by Chen et al. (2010, 2014). Campbell and Shiller (1987) demonstrate that in a present-value relationship, Granger causality and economic causality may go in opposite directions: while commodity prices have an impact on the exchange rate of the commodity exporting country, the latter has predictive ability on the former.[2]

The Chilean peso is particularly attractive as a commodity-currency due to the relevance of copper in the country's economy. According to the Central Bank of Chile, copper represented 48.02% of total Chilean exports in 2019. Besides, Chile is a net oil-importer country with about 10% of total imports focused on purchases of crude oil and its derivatives. See Central Bank of Chile (2018).

A direct implication of the CCH for Chile would say that the Chilean peso might have the ability to Granger-cause copper returns.[3] In principle it is not clear at all what this theory might imply for the connection between the Chilean exchange rate and fuel returns, provided that the share of oil-related products in the Chilean trade balance is not particularly high. Nevertheless, given that oil-related products display an important co-movement with base metal prices, it is reasonable to expect evidence of Granger causality from the Chilean peso to these oil-related products. In fact, Fig. 1 shows that monthly returns of copper, crude oil (West Texas Intermediate, WTI) and heating oil fluctuate closely together.[4]

Interestingly, in this paper we show that the predictive ability of the Chilean peso cannot only be justified on the grounds of the two natural explanations mentioned before. This is so, because our exercises with a multilateral exchange rate provide results with small differences relative to our exercises with the nominal Chilean Peso. We arrive to a similar conclusion regarding the importance of the CCH channel in our results. This is so, because we show that the predictive ability of the Chilean peso is only mildly overshadowed by the presence of a contemporaneous copper return in our specifications, remaining both statistically and economically significant. If the most important predictive channel were copper prices, then the inclusion of a contemporaneous copper term should kill or at least reduce importantly the predictive ability of the Chilean Peso.

We also show another plausible predictive channel: volatility. We report an important contemporaneous negative correlation between fuels returns and WTI realized volatility altogether with an interesting ability of Chilean Peso to predict this WTI realized volatility. Thus, the combination of both effects is a new plausible predictive channel from the Chilean peso to fuel returns.

We also think that the use of the Chilean exchange rate as a predictor is novel and somewhat surprising, because Chile is not a relevant player in the oil international market and the share of oil-related products in the Chilean trade balance is not particularly high. We think we are making a contribution to the literature by showing strong predictability from the Chilean peso to some important oil related products in addition to crude oil itself.

Finally, we add several additional exercises including: an analysis of potential predictive asymmetries between depreciations and appreciations, an analysis of parameter instability and a comparison between the predictive ability of the Chilean peso relative to other commonly used commodity-currencies.

The rest of the paper is structured as follows. In Section 2 we conduct a brief review of the related literature. In Section 3 we present the data and forecasting models. In Section 4 we report our empirical results both for our in-sample and out-of-sample exercises. In Section 5 we consider some plausible explanations for our findings. Section 6 shows some extensions and robustness exercises, while in Section 7 we present a summary of our analysis and the concluding remarks.

2. Literature review

There is a vast literature dedicated to crude oil price forecasting due to the importance that this commodity has in the global economy. Public institutions such as government agencies, central banks and international organizations need reliable forecasts of future oil price movements for policy design. Besides, private companies also use these predictions for their hedging strategies and investment decisions.

It is less frequent, however, to find articles focused on oil-derivative products, although a couple of interesting exceptions are the papers by Baumeister et al. (2016) and Anderson et al. (2011) that place their attention on the gasoline market.

Part of the literature focus on predicting nominal oil prices while other part focus on real prices.[5] This brief review focuses on the former given that our interest is to forecast nominal returns.

The first step in our review aims at exploring whether the empirical evidence shows indication of predictability. In other words we would like to explore if oil prices are predictable or not. Alquist et al. (2013) report significant reductions in MSPE relative to the no-change forecast when using a forecasting model that extrapolate changes in industrial raw materials and metal prices. They argue that increments in non-oil commodity prices are associated with changes in the global demand for industrial commodities, which depends on global economic activity. In another article, Gillman and Nakov (2009) find that since 1973 the US inflation has had the ability to Granger cause oil prices. Funk (2018),

[1] In the present-value relationship, the nominal exchange rate is the discounted sum of its fundamentals:

$$s_t = \gamma \sum_{j=0}^{\infty} \psi^j E_t\left(f_{t+j} | I_t\right)$$

where f_t represents the fundamental variable or the linear combination of variables that determine the nominal exchange rate. The information contained in f_t consists on the relevant economic factors that affect the currency, such as exporting commodity prices, interest rates differentials, money supply, output and inflation rates.

[2] The commodity-currencies literature focuses on forecasting the prices of the exporting commodities and not on a basket of fundamental variables determining the exchange rate. The reason for this is that these economies are heavily dependent on their commodity exports. We also target individual commodity returns and not a general vector of fundamentals.

[3] Chen et al. (2010) and Pincheira and Hardy (2019) show evidence supporting this claim.

[4] Moreover, Tables A1 and A2 in the appendix show correlations between all the commodities in our database. All entries in the tables are positive, and they are in general relatively high, suggesting a tight connection between copper and fuel returns, which might partially explain the predictive relationship between the Chilean exchange rate and fuel prices.

[5] It is also possible to classify the relevant literature according to whether the empirical analysis is carried out with variables in levels or in either returns or differences. For instance, many of the forecasting exercises in Alquist et al. (2013) are performed in levels, whereas the papers by Karolak et al. (2020), Fernández (2020) and West and Wong (2014) use mainly specifications in differences. It is also possible to find papers using both approaches in different exercises. As it is clear from Tables 1A and 1B, our specifications in this paper are expressed in returns.

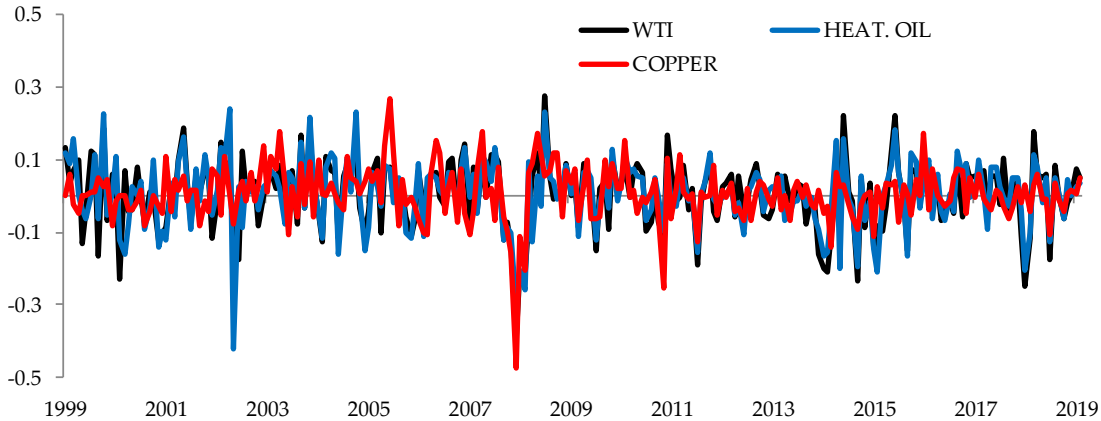

Fig. 1. Co-movement between copper and fuel returns.
Notes: Logarithmic monthly differences for three commodities: WTI, heating oil and copper. Sample period: 1999M11 – 2019M12. WTI and heating oil prices are from the Energy Information Administration database, while copper comes from the Chilean Copper Commission database. Source: Authors' elaboration.

Baumeister and Kilian (2015), Garratt et al. (2019) are other papers reporting predictability of oil prices mainly using forecast combination techniques. Therefore, according to these papers, predictability in the oil market is not a chimera.[6]

If oil prices are predictable, a second question to ask is how or what can predict oil prices. Natural potential leading indicators in the oil market are futures prices. While some empirical investigations suggest that the information contained in oil futures contracts is not very useful to forecast spot prices (see for instance Alquist et al., 2013; Alquist and Kilian, 2010; Chinn and Coibion, 2014; Fernández, 2017; and Knetsch, 2007), more recent research has shown promising results. For instance, Karolak et al. (2020), using ten years of weekly data, find that futures contracts outperform the driftless random walk benchmark when predicting WTI spot prices. Similarly, Bredin et al. (2021) show that models equipped with Nelson-Siegel factors constructed from the term structure curve of WTI futures prices are also able to outperform the no change benchmark in a forecasting competition. According to these recent papers, it seems natural to consider the information contained in futures to build a benchmark model.

It is reasonable to expect predictability from macroeconomic variables to oil prices as well, because global demand and economic activity have been historically the main factors behind crude oil price movements. Baumeister and Kilian (2016) study the determinants of the fluctuations in oil prices for the most relevant historical events. Traditionally, these instability periods were attributed to disruptions in global oil production due to exogenous political events occurred in OPEC countries, see Hamilton (2003). Baumeister and Kilian assess that, *au contraire*, demand shifts associated with the global business cycle are the most important factors behind major oil price fluctuations since 1973. Kilian and Murphy (2013), Hamilton (2009) and Kilian (2009) present more evidence in favor of this global business cycle argument.

In the quest for macroeconomic predictors, Kilian and Vega (2010) analyze the predictability of energy prices from thirty macroeconomic variables. They find no supportive evidence. Yet, Alquist et al. (2013) report predictability using commodity prices and some commodity-currencies as well. Similarly, Baumeister and Kilian (2012, 2015) construct a forecasting model based on a non-oil industrial raw materials index that demonstrates to have a strong predictive performance.

Closely related to our paper there is a literature exploring the interconnections between oil prices and exchange rates. For instance, Bützer et al. (2016) study the effect of three different oil price shocks (associated to global demand, oil-specific demand and oil supply) on the exchange rates of 43 advanced and emerging countries.[7] In addition, Gomez-Gonzalez et al. (2019) report in-sample evidence of bidirectional Granger causality between some exchange rates and commodity prices. Some other interesting contributions along these lines are those of Beckmann et al. (2017); Beckmann and Czudaj (2013); Brahmasrene et al. (2014); Akram (2009); and Lizardo and Mollick (2010).

Chen et al. (2010, 2014) present out-of-sample evidence of predictability from commodity exporters' exchange rates to a variety of globally traded primary products. Inspired in these papers, Alquist et al. (2013) explore predictability from the Australian and Canadian exchange rates to crude oil prices, finding out-of-sample evidence. Nevertheless, the literature that followed the influential empirical papers by Chen et al. (2010, 2014) has presented mixed evidence regarding the CCH. For instance, Chen et al. (2012) find more supportive evidence of the CCH using agricultural commodities. In the same line, Belasen and Demirer (2019) show results that are consistent with the CCH for a broad set of commodity exporters and exchange rates. Gargano and Timmermann (2014) also find more evidence of predictive ability using the Australian Dollar and the Indian Rupee, and Ciner (2017) show similar results using the South African Rand and white metals. In contrast, Lof and Nyberg (2017) use noncausal autoregressions to find that predictability from exchange rates to commodity prices disappears. Moreover, Groen and Pesenti (2011) find rather weak results of predictability using different commodity indices. All in all, the evidence regarding the CCH is somewhat mixed, and the scope and limits of its application are still not entirely clear. The literature also explores potential predictability not just for a single exporting commodity, but also, for other commodities that are closely related to it. This is explored in Pincheira and Hardy (2019), where they use the Chilean exchange rate to forecast the returns of the London Metal Exchange Index and of the six primary non-ferrous metals that are part of the index: aluminum, copper, lead, nickel, tin, and zinc. The authors report strong predictability, confirming that related commodities can also be forecast in the commodity-currencies framework. More recently, Pincheira and Hardy (2021a, 2021b) study the predictive relationship between six commodity-currencies (Australia, Canada, Chile, Iceland, New Zealand, and South Africa) and aluminum spot and futures returns. Notably, they show evidence of predictive ability even with the currencies of some commodity-exporting economies that have little or no production of aluminum at all, like South Africa or Chile. The authors argue that this predictive relationship relies on a strong co-movement between commodity prices.

[6] See Timmermann (2006) for an excellent review on forecast combinations.

[7] Chile is among these countries, although, the authors evaluate predictability from oil prices to exchange rates.

Table 1A
Econometric specifications one-step-ahead.

1: AR(1)	$\Delta \ln (FP_t) = c + \beta_1[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \rho \Delta \ln (FP_{t-1}) + \varepsilon_{1t}$
2: RW	$\Delta \ln (FP_t) = c + \beta_1[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \varepsilon_{2t}$
3: DRW	$\Delta \ln (FP_t) = \beta_1[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \varepsilon_{3t}$
4: Futures	$\Delta \ln (FP_t) = c + \beta_1[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \theta(Spread_{t-1}) + \varepsilon_{4t}$
5: Non-Oil	$\Delta \ln (FP_t) = c + \beta_1[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \psi_1[\Delta \ln (LMEX_{t-1}) + \ldots + \Delta \ln (LMEX_{t-3})] + \psi_2[\Delta \ln (LMEX_{t-4}) + \ldots + \Delta \ln (LMEX_{t-6})] + \varepsilon_{5t}$

Notes: FP_t stands for "fuel price at time t" while ER_t represents the relevant Exchange Rate at time t as well. For a generic variable x_t we define $\Delta \ln (x_t) \equiv \ln (x_t) - \ln (x_{t-1})$. $LMEX_t$ refers to the London Metal Exchange index at time t, while ε_{it} with $i = 1, \ldots, 5$ represent error terms. The variable $Spread_t$ is defined in Eq. (1) right next to Table 1A.
Source: Authors' elaboration.

While the co-movement argument is interesting and somewhat supported by these papers, in our view, it poses an interesting question about the limits of the CCH. In other words: Is it possible to forecast any commodity price using any commodity-currency, provided that commodity prices are correlated? Pincheira and Hardy (2021a, 2021b) partially explore this question and evaluate the predictive ability of six commodity-currencies over a wide variety of commodities, including not only base metals, but also bananas, lamb and wheat. The authors find poor and sporadic results: out of 432 out-of-sample exercises, they find evidence of predictive ability in only 8% of them. As pointed out by the authors, a predictive relationship from commodity-currencies to commodity prices should not be taken for granted. Accordingly, papers providing sound evidence of predictability are needed to identify the combination of commodity-currencies and commodity prices that display an important predictive linkage. In this sense, our paper contributes to this literature by showing a strong predictive relationship between the Chilean Peso and fuel returns, and also by showing a novel volatility channel that could be amplifying this predictability.

3. Data and methodology

3.1. Data

We use two reference measures for the case of crude oil: Brent prices and WTI delivered in Cushing, Oklahoma. We also consider three oil-derivative products: propane, regular gasoline and heating oil. In our empirical analysis we also make use of: copper prices, the London Metal Exchange (LME) index, the Chilean exchange rate defined as the amount of Chilean pesos required to buy one U.S. dollar, the Chilean Effective exchange rate, which is a multilateral index relative to a basket of relevant foreign currencies for Chile, the Australian Dollar, the Canadian Dollar and the Norwegian Krone.[8] Finally, we also consider WTI futures prices. In particular we pick the series labeled "futures contract 2" from the Energy Information Administration (EIA).

We have four different data sources. For the case of crude oil, refined products and WTI futures prices, the series are obtained from the EIA database. The LME index and all four exchange rates relative to the US dollar (Chile, Australia, Canada and Norway) come from Thomson Reuters Datastream. Copper prices come from the Chilean Copper Commission. Finally, the effective or multilateral Chilean exchange rate is extracted from the Central Bank of Chile database.

We downloaded daily prices for each asset, which we converted to monthly frequencies by sampling from the last observation of the month.[9] Table A3 and A4 in the appendix show some descriptive statistics.

The sample period goes from October 1999 to December 2019, resulting in 243 monthly observations of prices and 242 observations of one period returns. The starting point of our data coincides with the time when monetary authorities in Chile decided to pursue a pure flotation exchange rate regime.

3.2. Main forecasting models one-step-ahead

Our models for predictions one-step-ahead are quite simple. Following Pincheira and Hardy (2019), in our econometric specifications we use the six previous monthly returns of the Chilean exchange rate to predict fuel prices. These six lags of monthly returns are aggregated into two quarters, where each one is constructed by adding three subsequent individual monthly returns. Following this procedure, we have two parameters associated with the Chilean exchange rate in each model, one for each quarter.[10] We estimate five different models that are constructed by adding the Chilean peso to an oil-forecasting benchmark. These standard models include the AR(1) for commodities returns, the random walk (RW), the no-change model (DRW), a spread model based on crude oil futures prices and a last model considering the returns of non-oil commodities. The first three specifications come from the fact that auto-regressions and the random walk (with and without drift) are standard benchmarks to beat when predicting assets' returns.[11] The fourth model follows the tradition of the oil price forecasting literature by considering a specification based on crude oil future prices. Our fifth specification is based on the linkages between non-oil commodities and fuel prices that stems from the global demand for primary products. Consequently, we employ a benchmark model based on the LMEX index, which is an index of industrial metals. In additional exercises we explore the predictive ability of the Chilean Effective Exchange Rate, of the Australian and Canadian Dollar and of the Norwegian Krone. In those cases we consider the same specifications used for the Chilean exchange rate. We just replace the Chilean Peso for the relevant currency. In Table 1A we present our econometric specifications for predictions one-step-ahead.

For the fourth model we define:

$$Spread_{t-1} = ln\left(\frac{WTI_{t-1}^{F(t)}}{WTI_{t-1}}\right) \tag{1}$$

where $WTI_{t-1}{}^{F(t)}$ is the WTI futures price in time "t-1" for a contract with maturity in time t. Notice that the EIA database has four different types of WTI futures prices according to maturity. They are categorized from futures contract 1 through 4. The first contract has a delivery date within the next calendar month, so it means that it will expire before $t + 1$. The

[8] The LME index is constructed with the following weights: aluminum (42.8%), copper (31.2%), zinc (14.8%), lead (8.2%), nickel (2%) and tin (1%).

[9] By proceeding this way, we prevent our results from being polluted with serial autocorrelation, as it was first noticed in Working (1960).

[10] Basically, we use quarterly returns of the Chilean peso as predictors. This is useful because we can benefit from a reduction in the number of parameters in our models.

[11] For example, Chen et al. (2010, 2014) use the AR (1) and RW benchmarks, while in most articles analyzed in our review the no-change model plays an important role as a basic framework to compare more complex methods.

Table 1B
Econometric specifications multiple-steps-ahead.

6: AR(1)	$\Delta \ln (FP_{t-1,\ t-1+h}) = c^{(h)} + \beta_1{}^{(h)}[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2{}^{(h)}[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \rho^{(h)}\Delta \ln (P_{t-1}) + \varepsilon_{6t+h}$
7: RW	$\Delta \ln (FP_{t-1,\ t-1+h}) = c^{(h)} + \beta_1{}^{(h)}[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2{}^{(h)}[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \varepsilon_{7t+h}$
8: DRW	$\Delta \ln (FP_{t-1,\ t-1+h}) = \beta_1{}^{(h)}[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2{}^{(h)}[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \varepsilon_{8t+h}$
9: Futures	$\Delta \ln (FP_{t-1,\ t-1+h}) = c^{(h)} + \beta_1{}^{(h)}[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2{}^{(h)}[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \theta^{(h)}(Spread_{t-1}) + \varepsilon_{9t+h}$
10: Non-Oil	$\Delta \ln (FP_{t-1,\ t-1+h}) = c^{(h)} + \beta_1{}^{(h)}[\Delta \ln (ER_{t-1}) + \ldots + \Delta \ln (ER_{t-3})] + \beta_2{}^{(h)}[\Delta \ln (ER_{t-4}) + \ldots + \Delta \ln (ER_{t-6})] + \psi_1{}^{(h)}[\Delta \ln (LMEX_{t-1}) + \ldots + \Delta \ln (LMEX_{t-3})] + \psi_2{}^{(h)}[\Delta \ln (LMEX_{t-4}) + \ldots + \Delta \ln (LMEX_{t-6})] + \varepsilon_{10t+h}$

Notes: FP_t stands for "fuel price at time t" while ER_t represents the relevant Exchange Rate at time t as well. For a generic variable x_t we define $\Delta \ln (x_t) \equiv \ln (x_t) - \ln (x_{t-1})$ and $\Delta \ln (FP_{t-1,\ t-1+h}) \equiv \ln (FP_{t-1+h}) - \ln (FP_{t-1})$. $LMEX_t$ refers to the London Metal Exchange index at time t, while ε_{it+h} with $i = 6, \ldots, 10$ represent error terms. The variable $Spread_t$ is defined in Eq. (1) right next to Table 1A.
Source: Authors' elaboration.

second contract will not expire before $t+1$, but, it does not necessarily mature in that date exactly (normally, it matures after $t+1$). Contracts 3 and 4 have maturity dates at even longer horizons. We select contract 2 to build all our forecasts with futures, knowing that the matching is not perfect. In a similar context, this issue with the data is also documented in Alquist et al. (2013).

3.3. Main forecasting models multiple-steps-ahead

We use the direct method to build multistep-ahead-forecasts for horizons of h = 1, 3, 6, 9 and 12 months-ahead, see for instance, Karolak et al. (2020). We basically use the same specifications detailed in Table 1A, but adapted to obtain forecasts at multiple horizons. Table 1B shows these specifications.

In our in-sample and out-of-sample evaluations we test the following null hypothesis in specifications one-step-ahead:

$$H_0 : \beta_1 = \beta_2 = 0$$

Likewise, for specifications multiple-steps-ahead we consider the following null hypothesis

$$H_0^{(h)} : \beta_1^{(h)} = \beta_2^{(h)} = 0$$

With H_0 and $H_0{}^{(h)}$ we are testing if the information added by the relevant exchange rates is useful to improve the forecasts of the benchmark models. In our in-sample exercises we test the null hypotheses H_0 and $H_0{}^{(h)}$ using a traditional Wald statistic. In our out-of-sample evaluation we test H_0 either with the ENCNEW test proposed by Clark and McCracken (2001) or with the asymptotically normal test of Clark and West (2006, 2007). These tests are adequate tools to evaluate the null of equality in population MSPE in two nested models. At multiple horizons, we test $H_0{}^{(h)}$ using the test by Clark and West (2006, 2007).

In our in-sample analysis the parameters of the models are estimated using all the available observations. In our out-of-sample analysis we divide the total available sample in two windows: an initial estimation window of R observations and a prediction window of size P. This results in $R + P = T$, where T is the total number of observations in our sample. We select two different values for the ratio P/R for robustness. First, we employ one third of the observations for the initial estimation window, leaving two thirds for forecast evaluation. Second, we use approximately 60% of the observations for the initial estimation window, leaving 40% for evaluation.[12] We update the estimates of our parameters in each expanding estimation window. Please see West (2006) for additional details about out-of-sample evaluations in nested environments.

Tables 1A and 1B present our main forecasting specifications. In some other exercises, however, we use other specifications as well. In particular, in Section 5 we also show results when forecasting oil price volatility and in Section 6 we explore asymmetries in the predictive ability of the Chilean peso. These particular specifications are detailed either in the respective sections or in the Companion Appendix.

4. Empirical results

4.1. In-sample analysis

In Tables 2-6 we present selected results from our in-sample estimates of the models from Tables 1A-1B. Throughout this paper, we use HAC standard errors according to Newey and West (1987, 1994). To save space, here we only report 5 tables: those corresponding to the Non-oil benchmark at h = 1, 3, 6, 9 and 12 months ahead. One-step-ahead results for the rest of the specifications are available in the Companion Appendix. For the rest of the forecasting horizons tables are available upon request.

Tables 2-6 show an outstanding forecasting ability of the Chilean peso. In particular, these tables show that most of the coefficients associated with both distributed lags of the Chilean exchange rate are statistically significant at the 10% level for all fuels. This is particularly perceptible at the shortest horizons of one and three-months. Notice also that there is a tendency of the coefficient associated to the most distant distributed lag of the Chilean peso to lose relevance with the forecasting horizon. In the last row of Tables 2-6 we also present the Wald statistic p-value when the null hypothesis is that both coefficients associated to the Chilean exchange rate are zero. This null is rejected at the 10% significance level for all fuels in all tables. There is also a fair amount of cases in which the null is rejected at the 1% level. As to the coefficients of determination, there is an interesting "U-shape" pattern in Tables 2–3. By this we mean that the highest coefficient of determination is always achieved for WTI while the lowest is always achieved for Propane. For Gasoline, the coefficients of determination are higher than those of Propane. Depending on each table, Brent or Heating oil gets the second highest coefficient of determination. This pattern, however, tends to disappear at the longer horizons of 6, 9 and 12 months (Tables 4, 5 and 6).

Tables 2–6 show, in the great majority of the cases, that the sign of the coefficients associated with the distributed lags of the Chilean peso is negative. A similar finding is reported in Pincheira and Hardy (2019) when predicting copper prices with the Chilean exchange rate. The key difference between these two exercises is that Chile is a major copper exporter while a net oil importer. Negative coefficients associated to the Chilean peso are easy to explain in the copper forecasting exercise using a trade channel: if copper is expected to increase, then the Chilean exchange rate goes down in anticipation to futures inflows of US dollars into the Chilean economy. The same line of argument would suggest positive coefficients associated to the Chilean peso in the fuel forecasting exercise, yet most tables show negative coefficients as well, just like in the copper forecasting exercise in Pincheira and Hardy (2019). A plausible explanation for our findings relies on two well-known effects in the market of commodities: a "dollar effect" (See Akram, 2009) and a co-movement effect. The first effect is related to the fact that fuel prices

[12] For the first case ($P/R = 2$), $R = 81$ and $P = 162$. For the second case ($P/R = 0.6$), $R = 151$ and $P = 92$.

Table 2
Forecasting fuel returns one-month-ahead with the Chilean exchange rate.

In-sample analysis with specification (5) from Table 1A: The Non-Oil benchmark					
	WTI	Brent	Propane	Gasoline	Heating oil
ER(−1) + ER(−2) + ER(−3)	−0.303***	−0.304**	−0.306**	−0.336***	−0.233**
	(0.112)	(0.119)	(0.130)	(0.126)	(0.113)
ER(−4) + ER(−5) + ER(−6)	−0.516***	−0.614***	−0.422*	−0.582**	−0.474***
	(0.150)	(0.211)	(0.234)	(0.273)	(0.164)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.057	0.043	0.030	0.013	0.076
	(0.045)	(0.050)	(0.064)	(0.052)	(0.047)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.193***	−0.230***	−0.091	−0.232**	−0.149***
	(0.063)	(0.080)	(0.072)	(0.107)	(0.056)
Constant	0.008**	0.010**	0.003	0.009***	0.008*
	(0.004)	(0.004)	(0.006)	(0.003)	(0.004)
Observations	236	236	236	236	236
R-squared	0.141	0.128	0.044	0.078	0.119
Wald test p-value	0.001	0.007	0.010	0.013	0.007

Notes: ER stands for Chilean Exchange Rate Returns. WTI, Brent, Propane, Gasoline and Heating Oil represent one-month returns of each fuel. LMEX denotes London Metal Exchange index returns. Table 2 presents estimates of Eq. (5) in Table 1A. HAC standard errors are computed according to Newey and West (1987, 1994). *$p <$ 10%. **$p <$ 5%. ***p $<$ 1%.
Source: Authors' elaboration.

Table 3
Forecasting fuel returns three-months-ahead with the Chilean exchange rate.

In-sample analysis with specification (5) from Table 1B: The Non-Oil benchmark					
	WTI3	Brent3	Propane3	Gasoline3	Heating Oil3
ER(−3) + ER(−4) + ER(−5)	−1.166***	−1.201***	−0.757**	−1.278***	−0.986***
	(0.281)	(0.287)	(0.372)	(0.308)	(0.275)
ER(−6) + ER(−7) + ER(−8)	−1.127***	−1.168***	−0.709*	−1.187***	−1.057***
	(0.310)	(0.340)	(0.403)	(0.388)	(0.309)
LMEX(−3) + LMEX(−4) + LMEX(−5)	−0.056	−0.127	0.061	−0.279**	−0.015
	(0.128)	(0.140)	(0.161)	(0.139)	(0.116)
LMEX(−6) + LMEX(−7) + LMEX(−8)	−0.465***	−0.460***	−0.161	−0.389**	−0.330***
	(0.127)	(0.136)	(0.146)	(0.176)	(0.114)
Constant	0.023*	0.025*	0.005	0.025**	0.023*
	(0.013)	(0.013)	(0.018)	(0.012)	(0.012)
Observations	234	234	234	234	234
R-squared	0.231	0.201	0.072	0.136	0.202
Wald test p-value	0.000	0.000	0.068	0.000	0.000

Notes: ER stands for Chilean Exchange Rate Returns. WTI3, Brent3, Propane3, Gasoline3 and Heating Oil3 represent three-month returns of each fuel. LMEX denotes London Metal Exchange index one-month returns. Table 3 presents estimates of Eq. (5) in Table 1B. HAC standard errors are computed according to Newey and West (1987, 1994). *$p <$ 10%. **$p <$ 5%. ***p $<$ 1%.
Source: Authors' elaboration.

are denominated in US dollars. This implies that a higher value of this currency leads to a decrease in fuels' demand and a fall in their prices. The second effect simply indicates that commodity returns are positively correlated as can be seen in Table A2, which shows that the correlation between copper and fuel returns is positive and substantial. If the Chilean peso is a good predictor of copper returns, it might as well be a good predictor of commodities that are highly correlated with copper, like our sample of fuels.

Table 5 in the Online Appendix reports Wald tests *p*-values when testing the joint significance of both coefficients associated to the Chilean Peso in all our specifications, fuels and horizons. In the vast majority of the cases the null hypothesis is rejected at usual significance levels (rejection in 93% of the entries). At the shortest horizons of h = 1, 3 and 6 months the null is always rejected at least at the 10% significance level. In a few occasions at the longest horizons, however, the null cannot be rejected.

In summary, our in-sample estimates of econometric specifications from Table 1A and 1B show that the Chilean exchange rate has a strong ability to forecast fuel prices at short horizons of h = 1, 3 and 6 months. At horizons of h = 9 and h = 12 months there is still evidence of predictability, but there are also some exceptions with no evidence of predictability. The general picture suggests a slightly weaker predictive ability at these longer horizons.

In the next subsections we will see if these in-sample results hold true in the context of an out-of-sample analysis.

4.2. Out-of-sample analysis

Here we analyze predictability out-of-sample at the population level and at multiple horizons of h = 1, 3, 6, 9 and 12 months. Inference is carried out with the widely used test of Clark and West (2006, 2007) (henceforth CW test). Tables 7 and 8 show Root Mean Squared Prediction Errors (RMSPE) ratios between forecasts from the models including the Chilean Peso and the nested benchmarks. Ratios lower than 1 favor the model with the Chilean Peso, whereas ratios greater than 1 favor the benchmark model. Given that the CW test evaluates differences in MSPE at the population level, it still might reject the null when the ratios shown in Tables 7 and 8 are greater than 1. These paradoxical results are now well known in the forecasting literature and are explained in detail in the same papers by Clark and West (2006, 2007) and in many others, see for instance, West (2006) and Pincheira and West (2016).

Table 7 shows results when the number of one-step-ahead forecasts (P) is twice the number of observations used in the first estimation window (R), so $P/R = 2$. Table 8 shows results when the ratio $P/R = 0.6$. As mentioned before, we use an expanding scheme to update the estimates of our parameters.

Table 4
Forecasting fuel returns six-months-ahead with the Chilean exchange rate.

In-sample analysis with specification (5) from Table 1B: The Non-Oil benchmark					
	WTI6	Brent6	Propane6	Gasoline6	Heating Oil6
ER(−6) + ER(−7) + ER(−8)	−2.418***	−2.479***	−1.711***	−2.403***	−2.21***
	(0.533)	(0.549)	(0.612)	(0.517)	(0.526)
ER(−9) + ER(−10) + ER(−11)	−0.969**	−0.911*	−1.577***	−0.606	−1.268***
	(0.402)	(0.464)	(0.518)	(0.489)	(0.387)
LMEX(−6) + LMEX (−7) + LMEX (−8)	−0.619***	−0.700***	−0.288	−0.721***	−0.482***
	(0.22)	(0.252)	(0.221)	(0.257)	(0.157)
LMEX(−9) + LMEX (−10) + LMEX (−11)	−0.359**	−0.281	−0.326	−0.187	−0.273
	(0.171)	(0.196)	(0.273)	(0.184)	(0.182)
Constant	0.039	0.043	0.012	0.038	0.042
	(0.026)	(0.029)	(0.039)	(0.026)	(0.027)
Obs	231	231	231	231	231
R-squared	0.216	0.194	0.120	0.166	0.219
Wald test p-value	0.000	0.000	0.000	0.000	0.000

Notes: ER stands for Chilean Exchange Rate Returns. WTI6, Brent6, Propane6, Gasoline6 and Heating Oil6 represent six-month returns of each fuel. LMEX denotes London Metal Exchange index one-month return. Table 4 presents estimates of Eq. (5) in Table 1B. HAC standard errors are computed according to Newey and West (1987, 1994). *$p < 10\%$. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

Table 5
Forecasting fuel returns nine-months-ahead with the Chilean exchange rate.

In-sample analysis with specification (5) from Table 1B: The Non-Oil benchmark					
	WTI9	Brent9	Propane9	Gasoline9	Heating Oil9
ER(−9) + ER(−10) + ER(−11)	−2.038***	−2.001***	−2.343***	−1.633***	−2.233***
	(0.448)	(0.481)	(0.596)	(0.480)	(0.425)
ER(−12) + ER (−13) + ER (−14)	−0.462	−0.192	−1.158	0.041	−0.607
	(0.817)	(0.872)	(0.978)	(0.763)	(0.726)
LMEX(−9) + LMEX (−10) + LMEX (−11)	−0.607***	−0.610**	−0.472	−0.574**	−0.498**
	(0.229)	(0.261)	(0.333)	(0.245)	(0.194)
LMEX(−12) + LMEX(−13) + LMEX(−14)	−0.131	0.025	−0.187	0.052	0.033
	(0.325)	(0.37)	(0.475)	(0.329)	(0.332)
Constant	0.042	0.046	0.01	0.039	0.046
	(0.044)	(0.047)	(0.061)	(0.042)	(0.044)
Obs	228	228	228	228	228
R-squared	0.100	0.084	0.099	0.065	0.125
Wald test p-value	0.000	0.000	0.000	0.001	0.000

Notes: ER stands for Chilean Exchange Rate Returns. WTI9, Brent9, Propane9, Gasoline9 and Heating Oil9 represent nine-month returns of each fuel. LMEX denotes London Metal Exchange index one-month return. Table 5 presents estimates of Eq. (5) in Table 1B. HAC standard errors are computed according to Newey and West (1987, 1994). *$p < 10\%$. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

At the population level, Tables 7 and 8 show an outstanding predictive performance of the Chilean Peso for all five fuels and benchmarks at the horizons of h = 1, 3, 6 and 9 months, as the CW test rejects the null in almost every corresponding entry in both tables. This is consistent with our in-sample results shown in Tables 2-6 and is also consistent with the good out-of-sample results reported in Tables 6 and 7 in the Companion Appendix.[13]

At the longest horizon of h = 12, results are slightly weaker, and we see several entries in both tables in which it is not possible to reject the null of equality in population MSPE. Moreover at this forecasting horizon both tables report several entries with RMSPE ratios greater than 1, which also indicates a relatively poor forecasting performance of the models containing the Chilean Peso at the sample level.

At shorter horizons this performance is better, with most of the RMSPE ratios lower than one and sometimes by a relatively important margin. For instance, when the benchmark is the driftless random walk and h = 6, Tables 7–8 show ratios lower than 0.95 for WTI and Heating Oil. The lowest ratio (0.89) is achieved for Heating Oil in Table 8, for h = 6, when the benchmark is the model predicting with futures.

In summary, we see an outstanding predictive performance of the Chilean peso at horizons of h = 1, 3, 6, and 9 months when this performance is measured at the population level. Results at the sample level are moderate but consistent across both Tables 7 and 8 favoring the models including the Chilean Peso.

The CW test is designed to compare population MSPE of two nested models. In this framework, we show that the Chilean peso does a remarkable job at reducing the population MSPE of our benchmark models. A specification that yields lower population MSPE relative to a benchmark means that it has superior predictive ability. As mentioned before, population MSPE based on true parameters may be quite different from their finite sample counterparts. This is consistent with our results in Tables 7 and 8 which show a remarkable predictive ability for the Chilean peso at a population level, but rather a moderate performance when the model is put into practice with estimates of the actual unknown population parameters. This might be frustrating for applied forecasters that are interested in the forecasts actually available for them: those constructed with estimates of the true population parameters.

Pincheira and Hardy (2021a, 2021b) pointed out that the MSPE by

Table 6
Forecasting fuel returns one-year-ahead with the Chilean exchange rate.

In-sample analysis with specification (5) from Table 1B: The Non-Oil benchmark					
	WTI12	Brent12	Propane12	Gasoline12	Heating Oil12
ER(−12) + ER(−13) + ER(−14)	−1.490*	−1.299	−1.687*	−0.953	−1.568**
	(0.841)	(0.900)	(0.987)	(0.796)	(0.724)
ER(−15) + ER(−16) + ER(−17)	−0.074	−0.037	0.078	0.251	−0.156
	(1.040)	(1.086)	(1.226)	(0.932)	(1.030)
LMEX(−12) + LMEX (−13) + LMEX(−14)	−0.304	−0.246	−0.169	−0.23	−0.135
	(0.396)	(0.454)	(0.527)	(0.402)	(0.337)
LMEX(−15) + LMEX (−16) + LMEX(−17)	−0.035	0.049	−0.004	0.085	0.097
	(0.415)	(0.445)	(0.503)	(0.404)	(0.422)
Constant	0.045	0.053	0.001	0.043	0.051
	(0.056)	(0.061)	(0.076)	(0.054)	(0.057)
Obs	225	225	225	225	225
R-squared	0.048	0.032	0.041	0.022	0.062
Wald test p-value	0.006	0.027	0.015	0.097	0.007

Notes: ER stands for Chilean Exchange Rate Returns. WTI12, Brent12, Propane12, Gasoline12 and Heating Oil12 represent one-year-returns of each fuel. LMEX denotes London Metal Exchange index one-month return. Table 6 presents estimates of Eq. (5) in Table 1B. HAC standard errors are computed according to Newey and West (1987, 1994). *$p < 10\%$. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

[13] Tables 6 and 7 in the Companion Appendix present results of the ENCNEW test of Clark and McCracken (2001) for the econometric specifications from Table 1A and for the two different divisions of our sample. These tables show an outstanding forecasting ability of the Chilean peso relative to all our benchmarks and for all of our commodities. For WTI, Brent and heating oil, the null of no predictability of the Chilean exchange rate is rejected at the 1% significance level in both tables. For Gasoline, rejection of the same null is also obtained at the 1% level with only one exception in which rejection is achieved only at the 5% significance level. For Propane our results are less compelling, but again, rejection of the null is achieved in every entry of Tables 6 and 7 at least at the 10% significance level.

Table 7
Out-of-sample RMSPE ratios, P/R = 2.

Inference is carried out with the test by Clark and West (2007)					
	WTI	Brent	Propane	Gasoline	Heating oil
Benchmark: AR(1)					
h = 1	0.958*	0.949**	0.98**	0.967**	0.935**
h = 3	0.923**	0.942**	0.959**	0.957**	0.891**
h = 6	0.922**	0.930**	0.954**	0.925***	0.92***
h = 9	0.984*	0.989*	0.964**	1.000	0.959**
h = 12	1.012	1.020	1.000	1.024	0.996
Benchmark: RW					
h = 1	0.944**	0.945**	0.981**	0.973*	0.933**
h = 3	0.929**	0.953**	0.968**	0.983*	0.903**
h = 6	0.933***	0.947***	0.964**	0.961**	0.929***
h = 9	0.996*	1.002*	0.972**	1.024	0.969*
h = 12	1.020	1.029	1.004	1.034	1.000
Benchmark: DRW					
h = 1	0.946**	0.947**	0.980**	0.974*	0.936**
h = 3	0.934**	0.958**	0.970**	0.990*	0.908**
h = 6	0.938***	0.954***	0.966**	0.970**	0.936***
h = 9	1.006*	1.011*	0.979*	1.034	0.985**
h = 12	1.030	1.036	1.010	1.041	1.015
Benchmark: Futures					
h = 1	0.936**	0.939**	0.981**	0.970*	0.933**
h = 3	0.918**	0.945**	0.963**	0.974**	0.894**
h = 6	0.930***	0.944***	0.954**	0.957**	0.927***
h = 9	0.988*	0.996*	0.948**	1.020	0.957**
h = 12	1.010	1.021	0.982*	1.028	0.985
Benchmark: Non-Oil					
h = 1	0.960***	0.946***	0.999*	0.974**	0.956***
h = 3	0.921***	0.928***	0.995*	0.949***	0.914***
h = 6	0.911***	0.921***	0.964**	0.932***	0.906***
h = 9	0.983**	0.990**	0.985*	1.004**	0.964**
h = 12	1.022	1.025	1.013	1.027	1.013

Notes: Figures lower than 1 favor the models with the Chilean Peso. P represents the number of one-step-ahead forecasts and R is the sample size of the first estimation window. AR(1), RW, DWR, FUTURES and NON-OIL benchmarks correspond to models 1,2,3,4 and 5 from Table 1A and Table 1B when the coefficients associated to the Chilean peso are zero. *$p < 10\%$. **$p < 5\%$. ***p < 1%.
Source: Authors' elaboration.

Table 8
Out-of-sample RMSPE ratios, P/R = 0.6.

Inference is carried out with the test by Clark and West (2007)					
	WTI	Brent	Propane	Gasoline	Heating oil
Benchmark: AR(1)					
h = 1	0.981**	0.983**	0.999	0.980**	0.973**
h = 3	0.940***	0.938***	0.979*	0.945**	0.914***
h = 6	0.906***	0.918***	0.942**	0.914***	0.897***
h = 9	0.983***	0.993***	0.956**	1.004*	0.943***
h = 12	0.980***	0.990**	0.980**	1.000	0.955***
Benchmark: RW					
h = 1	0.964**	0.966**	0.992*	0.982**	0.966**
h = 3	0.945**	0.944**	0.980*	0.962*	0.926**
h = 6	0.920***	0.934***	0.945**	0.948**	0.902***
h = 9	0.998**	1.007	0.954**	1.030	0.947***
h = 12	0.990***	1.000	0.982*	1.016	0.957***
Benchmark: DRW					
h = 1	0.971**	0.973**	0.993*	0.986*	0.976**
h = 3	0.956**	0.956**	0.983*	0.974*	0.943**
h = 6	0.930***	0.946**	0.950**	0.962**	0.922***
h = 9	1.006	1.014	0.959**	1.041	0.965***
h = 12	0.990***	0.997*	0.985*	1.016	0.962***
Benchmark: Futures					
h = 1	0.957**	0.963**	0.990*	0.981**	0.964**
h = 3	0.928**	0.932**	0.973*	0.946**	0.916**
h = 6	0.908***	0.924***	0.924**	0.942**	0.889***
h = 9	0.987**	0.997**	0.921**	1.021	0.931***
h = 12	0.980***	0.989***	0.958**	1.005	0.942***
Benchmark: Non-Oil					
h = 1	0.955***	0.958***	0.992*	0.974***	0.969***
h = 3	0.913***	0.918***	0.978**	0.931***	0.911***
h = 6	0.889***	0.897***	0.929***	0.921***	0.890***
h = 9	0.972**	0.982**	0.944**	1.006	0.950**
h = 12	0.992**	1.001	1.001	1.010	0.982**

Notes: Figures lower than 1 favor the models with the Chilean Peso. P represents the number of one-step-ahead forecasts and R is the sample size of the first estimation window. AR(1), RW, DWR, FUTURES and NON-OIL benchmarks correspond to models 1,2,3,4 and 5 from Table 1A and Table 1B when the coefficients associated to the Chilean peso are zero. *$p < 10\%$. **$p < 5\%$. ***p < 1%.
Source: Authors' elaboration.

itself does not always provide a complete assessment of the predictive ability of a given forecast. The authors show that under some conditions of inefficiency, the forecast displaying the lowest MSPE may also display the lowest correlation with the target variable. They label this phenomenon as "The MSPE Paradox." In this sense, Pincheira and Hardy (2021a, 2021b) propose to use correlations as a complementary tool to evaluate the performance of a given forecast. To this end, we explore the correlation between out-of-sample forecasts and the target variable, which is nothing but actual returns of the corresponding fuel. In Table 9 next we present these correlations for the specific case of crude oil WTI.

Table 9 reports positive correlations between one-step-ahead forecasts and actual WTI returns. Higher correlations are detected when we use one-third of the sample as the first expanding window. Although moderate, all of the reported correlations are in the 0.20–0.36 range, which intuitively seems high enough to be useful guidance for investors, policy makers, and practitioners.

All across this paper, we have shown sound evidence of predictability from the Chilean exchange rate to fuel prices, but we also explored the potential predictive relationship in the opposite direction: that is, from fuels to the Chilean peso. Our in-sample and out-of-sample results were really poor, basically indicating neither systematic nor substantial evidence of predictability in this direction. These results are consistent with those reported by Chen et al. (2010) and Pincheira and Hardy (2019). They are available upon request.

Table 9
Forecasting WTI Returns One-Month-Ahead with the Chilean Exchange Rate
Correlations between forecasts generated with specifications from Table 1A and actual WTI returns.

	P/R=2	P/R=0.6
	Correlations	Correlations
AR(1)	0.30	0.22
RW	0.31	0.22
DRW	0.33	0.23
FUTURES	0.35	0.31
NON-OIL	0.36	0.29

Notes: AR(1), RW, DRW, FUTURES, and NON-OIL benchmarks correspond to models 1,2,3,4, and 5 from Table 1A.
Source: Authors' elaboration.

5. Plausible explanations

Thus far, we have argued that predictability from the Chilean peso to fuel returns may be the consequence of the CCH, which implies predictability from the Chilean Exchange Rate to copper prices, plus a strong co-movement between copper and fuel prices. We have also mentioned that our results can be explained by a "dollar effect." We explore next the relevance of this last effect.

5.1. The dollar effect

To explore the importance of a potential "dollar effect," we repeat all our predictive exercises, but instead of using the Chilean Peso as a predictor, we use the Chilean Effective Exchange Rate. This rate is a multilateral index that represents the nominal value of the Chilean Peso relative to a basket of 28 foreign currencies. The shares of each foreign exchange rates are computed according to the relevance of the respective country in the Chilean trade balance. Data for the Chilean Effective Exchange Rate are available at the Central Bank of Chile on a daily frequency. We build a monthly series by sampling from the last day of the month.

In Table 10 we present in-sample estimates of the model corresponding to the Non-oil benchmark from Table 1A, when forecasting with the Chilean Effective Exchange Rate (ECLP). One-step-ahead results for the rest of the specifications are available in the Companion Appendix (Tables 8–11). In the same appendix, we present in-sample estimates for the Non-oil benchmark but at longer horizons of h = 3, 6, 9, and 12 months as well. See Tables 12–15 in the Companion Appendix.

Relative to the tables constructed with the nominal Chilean exchange rate, Tables 10 and Tables 12–15 in the Companion Appendix are very similar. In particular, we see that most coefficients associated to the distributed lags of the ECLP are statistically significant and negative, which is exactly the same that happens in Tables 2-6. Moreover, coefficients of determination are again similar to those in Tables 2-6 and the Wald tests rejects, in all cases, the null that evaluates the joint significance of the coefficients associated to the ECLP, just as they do in Tables 2-6. So, in summary, our in-sample results with the ECLP tell basically the same story as Tables 2-6.

In Table 16 in the Companion Appendix, we present the *p*-values of the Wald test evaluating the null of no predictive ability of the ECLP, just like we do in Table 5 in the same Appendix with the Chilean Peso. Results are again similar, as there are only five *p*-values indicating no rejection of the null hypothesis at usual significance levels.

All in all, our in-sample analyses with the ECLP provide the same signal as our exercises with the Chilean Peso: there is strong evidence of predictability from the Chilean currency to fuel returns. Consequently, these results indicate that this predictability goes beyond a simple "dollar effect."

Tables 11–12 show the results of our out-of-sample exercises with the ECLP. They are akin to Tables 7–8 in Section 4 and therefore show RMSPE ratios and the results of the CW test.

Out-of-sample results with the ECLP are again similar to those obtained with the nominal Chilean peso. In particular, most RMSPE ratios in Tables 11–12 are lower than 1 and many entries of the tables report statistically significant results in favor of the models with the ECLP. We note one interesting difference relative to the tables with the Chilean Peso: results when P/R = 0.6 are slightly weaker when forecasting with the ECLP, which indicates that a "dollar effect" could be somewhat more important when forecasting in the last share of our sample. This is particularly perceptible for Propane. Despite this difference, predictability with the ECLP is strong in many commodities and at several horizons, indicating that the predictive ability reported for the Chilean Peso goes beyond a simple "dollar effect."

5.2. CCH plus comovement

Predictability from the Chilean peso to fuel returns may be the consequence of the CCH, which implies predictability from the Chilean Exchange Rate to copper prices, plus a strong co-movement between copper and fuel prices. This strong co-movement could be modeled with the following simple equation:

Table 10
Forecasting fuel returns one-month-ahead with the Chilean effective exchange rate.

In-sample analysis with specification (5) from Table 1A: The Non-Oil benchmark					
	WTI	Brent	Propane	Gasoline	Heating Oil
ECLP(−1) + ECLP(−2) + ECLP(−3)	−0.314***	−0.311**	−0.374*	−0.339**	−0.269**
	(0.116)	(0.123)	(0.192)	(0.145)	(0.123)
ECLP(−4) + ECLP(−5) + ECLP(−6)	−0.541***	−0.612**	−0.323	−0.622*	−0.444**
	(0.197)	(0.284)	(0.300)	(0.341)	(0.203)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.099**	0.09**	0.068	0.061	0.108***
	(0.047)	(0.042)	(0.058)	(0.043)	(0.041)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.133**	−0.155**	−0.014	−0.168*	−0.082*
	(0.054)	(0.067)	(0.059)	(0.086)	(0.046)
Constant	0.004	0.005	−0.001	0.004	0.004
	(0.005)	(0.005)	(0.006)	(0.004)	(0.005)
Observations	236	236	236	236	236
R-squared	0.136	0.117	0.038	0.075	0.108
Wald test p-value	0.001	0.01	0.023	0.006	0.012

Notes: ECLP stands for Effective Chilean Exchange Rate Returns. WTI, Brent, Propane, Gasoline and Heating Oil represent one-month returns of each fuel. LMEX denotes London Metal Exchange index returns. Table 10 presents estimates of Eq. (5) in Table 1A. *$p < 10\%$. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

Table 11
Out-of-sample RMSPE ratios for the Chilean effective exchange rate P/R = 2.

Inference is carried out with the test by Clark and West (2007)					
	WTI	Brent	Propane	Gasoline	Heating oil
Benchmark: AR(1)					
h = 1	0.963*	0.954*	0.998*	0.967*	0.948**
h = 3	0.906**	0.919**	0.972*	0.932**	0.882**
h = 6	0.916**	0.923**	0.940**	0.916**	0.908**
h = 9	0.973*	0.976*	0.948**	0.981	0.946**
h = 12	0.990*	0.997*	0.985**	1.002*	0.978*
Benchmark: RW					
h = 1	0.947*	0.948**	0.998*	0.972**	0.944**
h = 3	0.908**	0.927**	0.978*	0.956**	0.890**
h = 6	0.922**	0.934**	0.946**	0.943**	0.912**
h = 9	0.977*	0.981*	0.951**	0.994	0.949**
h = 12	0.993*	1.001	0.987*	1.009	0.979*
Benchmark: DRW					
h = 1	0.947*	0.947**	0.998*	0.972**	0.944**
h = 3	0.908**	0.928**	0.979*	0.960**	0.889**
h = 6	0.923***	0.937**	0.95***	0.948**	0.914***
h = 9	0.983*	0.986*	0.959**	1.001	0.960**
h = 12	0.999*	1.005	0.994**	1.012	0.989*
Benchmark: Futures					
h = 1	0.944*	0.946**	0.999*	0.972*	0.946**
h = 3	0.910**	0.931**	0.978*	0.956**	0.891**
h = 6	0.930**	0.940**	0.946**	0.947**	0.919**
h = 9	0.976	0.980	0.939**	0.994	0.946**
h = 12	0.993	1.002	0.974**	1.008	0.976**
Benchmark: Non-Oil					
h = 1	0.975***	0.968***	1.021	0.986*	0.977***
h = 3	0.929***	0.934***	1.001	0.952***	0.922***
h = 6	0.928***	0.936***	0.954***	0.943**	0.912***
h = 9	0.978**	0.982*	0.963**	0.989*	0.952***
h = 12	1.000	1.005	0.995*	1.011	0.992*

Notes: Figures lower than 1 favor the models with the Chilean Effective Exchange Rate. P represents the number of one-step-ahead forecasts and R is the sample size of the first estimation window. AR(1), RW, DWR, FUTURES and NON-OIL benchmarks correspond to models 1,2,3,4 and 5 from Table 1A and Table 1B when the coefficients associated to the Chilean peso are zero. *$p < 10\%$. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

Table 12
Out-of-sample RMSPE ratios for the Chilean effective exchange rate P/R = 0.6.

Inference is carried out with the test by Clark and West (2007)					
	WTI	Brent	Propane	Gasoline	Heating oil
Benchmark: AR(1)					
h = 1	0.999*	1.008	1.011	0.997*	0.996*
h = 3	0.944***	0.949***	1.001	0.961**	0.932**
h = 6	0.939***	0.95***	0.976*	0.956**	0.95**
h = 9	0.991*	0.997	0.954**	1.004	0.966*
h = 12	0.979**	0.989*	0.972**	0.991*	0.966**
Benchmark: RW					
h = 1	0.982**	0.985**	1.004	0.995*	0.988*
h = 3	0.946**	0.950**	0.998	0.968*	0.939**
h = 6	0.941**	0.953**	0.974*	0.970**	0.945**
h = 9	0.996	1.001	0.948**	1.018	0.961**
h = 12	0.984**	0.992**	0.972**	1.001	0.963***
Benchmark: DRW					
h = 1	0.985**	0.988*	1.005	0.998	0.993
h = 3	0.952**	0.956**	1.001	0.975*	0.947**
h = 6	0.948**	0.961**	0.980*	0.983*	0.958**
h = 9	1.002	1.005	0.955**	1.027	0.974*
h = 12	0.980**	0.985**	0.975**	0.997	0.964***
Benchmark: Futures					
h = 1	0.983*	0.987*	1.005	0.999	0.989*
h = 3	0.954**	0.958**	1.002	0.969*	0.946*
h = 6	0.952**	0.961**	0.970*	0.979*	0.954*
h = 9	0.995	0.999	0.935**	1.014	0.959*
h = 12	0.983**	0.990**	0.959**	0.996	0.960**
Benchmark: Non-Oil					
h = 1	0.977**	0.984**	1.007	0.992*	0.990*
h = 3	0.944***	0.953**	0.999	0.962***	0.942***
h = 6	0.946**	0.952**	0.972*	0.968**	0.961*
h = 9	0.992	0.994	0.951**	1.009	0.979
h = 12	0.987	0.993	0.985	0.997	0.983

Notes: Figures lower than 1 favor the models with the Chilean Effective Exchange Rate. P represents the number of one-step-ahead forecasts and R is the sample size of the first estimation window. AR(1), RW, DWR, FUTURES and NON-OIL benchmarks correspond to models 1,2,3,4 and 5 from Table 1A and Table 1B when the coefficients associated to the Chilean peso are zero. $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

$$\Delta ln\left(FP_{t,t+h}\right) = c + \rho\Delta ln\left(Copper_{t,t+h}\right) + v_{t+h} \tag{11}$$

where $\Delta \ln (X_{t,\ t+h}) \equiv \ln (X_{t+h}) - \ln (X_t)$ for a generic variable X_t, v_{t+h} is an orthogonal shock to accumulated Copper returns, FP_t represents a generic fuel price at time t and similarly, $Copper_t$ represents Copper price at time t.

If information contained in the Chilean peso were only able to predict copper returns, and not idiosyncratic components of fuel prices summarized in the shock v_{t+h} , we should not be able to reject the null

$$H_{00}^{(h)} : \beta_1^{(h)} = \beta_2^{(h)} = 0$$

in the following regression including $\Delta\ ln\ (Copper_{t,\ t+h})$:

$$\begin{aligned}\Delta ln\left(FP_{t,t+h}\right) =& c + \rho\Delta ln\left(Copper_{t,t+h}\right) + \beta_1^{(h)}\Delta ln\left(CLP_{t-k,t}\right) \\ &+ \beta_2^{(h)}\Delta ln\left(CLP_{t-k-l,t-k}\right) + u_{t+h}\end{aligned} \tag{12}$$

where $\Delta\ ln\ (CLP_{t-k,\ t})$ denotes the accumulated log returns of the Chilean peso in the time period from t-k to t, where k and l are positives integers. On the contrary, rejection of $H_{00}{}^{(h)}$ would mean that the Chilean peso has the ability to predict fuel returns beyond its own ability to predict copper prices. We explore these interesting null hypotheses by augmenting all our specifications in Tables 1A and 1B with a contemporaneous accumulated copper return as in expression (12). Table 17 in the Companion Appendix and Table 13 next provide a clear picture that is quite general across all our specifications.

It is interesting to compare results in Table 13 with those of Table 4. The only difference between them is the contemporaneous copper term. Its inclusion induces much higher coefficients of determination, lower coefficients associated to the Chilean peso (in absolute value) and lower standard errors as well. Our reading of these results is that a contemporaneous copper term reduces the ability of the Chilean Peso to predict fuel returns, yet, this predictability remains sizable and important.

Results in Tables 13 and 17 in the Companion Appendix are just two examples of a general feature: Lagged Chilean peso returns survive to the presence of a contemporaneous copper term in most of our 125 regressions including five forecasting horizons, five specifications and five fuels. We do not report the specific tables for all the specifications, but they are available upon request.

Table 14 reports out-of-sample results from exercises that are the natural extension of those reported in Tables 11 and 12. The only difference being the inclusion of a contemporaneous copper return term in all the specifications.

Most of the entries in Table 14 show statistically significant results in favor of the models containing distributed lags of the Chilean Peso, and also, most of the entries show RMSPE ratios below 1, which again favors the models with the Chilean currency. Predictability seems to be stronger for WTI, Brent and Heating Oil, but the main message we would like to emphasize here is that for many fuels, horizons and benchmarks, the predictive ability of the Chilean peso survives to the presence of a contemporaneous copper return. These results are consistent with those shown in Table 17 in the Companion Appendix and those shown in Table 13 here. They indicate that the predictive relationship between fuels and the Chilean Currency goes far beyond the Commodity Currency Hypothesis in combination with a co-movement argument.

5.3. The contemporaneous leverage effect

We have shown that the ability of the Chilean Peso to predict fuel returns goes beyond the "dollar effect" and a combination of the CCH and co-movement in commodity prices. The important question to address is: what else could be behind this strong predictive power? In

Table 13
Forecasting fuel returns six-months-ahead with the Chilean exchange rate and a contemporaneous copper term.

In-sample analysis with specification (5) from Table 1B: The Non-Oil benchmark					
	WTI6	*Brent6*	*Propane6*	*Gasoline6*	*Heating Oil6*
ER(−6) + ER(−7) + ER(−8)	−1.457***	−1.419***	−0.626	−1.377***	−1.309***
	(0.332)	(0.329)	(0.468)	(0.344)	(0.351)
ER(−9) + ER(−10) + ER(−11)	−0.545**	−0.443*	−1.098***	−0.153	−0.871***
	(0.249)	(0.267)	(0.343)	(0.327)	(0.244)
Copper	0.689***	0.760***	0.778***	0.736***	0.646***
	(0.124)	(0.138)	(0.123)	(0.101)	(0.119)
LMEX(−6) + LMEX (−7) + LMEX (−8)	−0.331**	−0.382***	0.038	−0.413***	−0.212
	(0.145)	(0.14)	(0.174)	(0.149)	(0.143)
LMEX(−9) + LMEX (−10) + LMEX (−11)	−0.101	0.004	−0.035	0.088	−0.031
	(0.085)	(0.086)	(0.182)	(0.121)	(0.105)
Constant	0.008	0.008	−0.024	0.004	0.012
	(0.021)	(0.021)	(0.031)	(0.018)	(0.022)
Obs	231	231	231	231	231
R-squared	0.555	0.574	0.411	0.510	0.546

Notes: ER stands for Chilean Exchange Rate Returns. WTI6, Brent6, Propane6, Gasoline6 and Heating Oil6 represent six-month returns of each fuel. LMEX denotes London Metal Exchange index one-month return. Copper represents a contemporaneous six- month copper return. Table 13 is based on Eq. (5) in Table 1B but augmented with a contemporaneous six-month copper return. HAC standard errors are computed according to Newey and West (1987, 1994). $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

Table 14
Out-of-sample RMSPE ratios when forecasting fuel returns multiple-steps-ahead with the Chilean exchange rate and a contemporaneous copper term.

Out-of-sample RMSPE ratios, P/R = 0.6						Out-of-sample RMSPE ratios, P/R = 2					
	WTI	Brent	Propane	Gasoline	Heating oil		WTI	Brent	Propane	Gasoline	Heating oil
Benchmark: AR(1)+Copper						Benchmark: AR(1)+Copper					
h = 1	0.982**	0.977***	0.999	0.985**	0.974**	h = 1	0.975**	0.963***	1.000**	0.982**	0.958***
h = 3	0.953***	0.947***	0.989	0.957**	0.921***	h = 3	0.944***	0.963***	0.995	0.968**	0.909***
h = 6	0.929***	0.929***	0.952**	0.928***	0.906***	h = 6	0.974***	0.992**	0.972*	0.964***	0.981*
h = 9	0.973***	0.977***	0.947**	0.989**	0.920***	h = 9	1.029	1.042	0.977*	1.047	1.016
h = 12	0.968***	0.972***	0.975*	0.977**	0.926***	h = 12	1.004	1.019	0.984*	1.002	0.989
Benchmark: RW + Copper						Benchmark: RW + Copper					
h = 1	0.972**	0.972***	0.995	0.989*	0.974**	h = 1	0.967**	0.967***	1.000***	0.989**	0.961***
h = 3	0.958**	0.959**	0.992	0.975*	0.94***	h = 3	0.950***	0.977***	1.001	1.001*	0.925***
h = 6	0.939***	0.947***	0.958*	0.961**	0.916***	h = 6	0.978***	1.004*	0.982*	0.999*	0.987*
h = 9	0.984**	0.990**	0.948*	1.013	0.926***	h = 9	1.037	1.051	0.987	1.079	1.028
h = 12	0.973**	0.979***	0.979	0.996	0.921***	h = 12	1.008	1.023	0.987*	1.017	0.989
Benchmark: DRW + Copper						Benchmark: DRW + Copper					
h = 1	0.974**	0.974***	0.995	0.990	0.978**	h = 1	0.967**	0.967***	0.999*	0.990**	0.963***
h = 3	0.96**	0.961**	0.991	0.977*	0.946**	h = 3	0.951***	0.977***	1.002	1.007	0.926***
h = 6	0.940***	0.949***	0.953**	0.963**	0.924***	h = 6	0.965***	0.99***	0.979*	1.003*	0.963**
h = 9	0.986**	0.991*	0.942**	1.016	0.939***	h = 9	1.004*	1.014**	0.969**	1.060	0.982**
h = 12	0.975**	0.978***	0.976*	1.000	0.932***	h = 12	0.986**	0.994	0.987*	1.012	0.962**
Benchmark: Futures+Copper						Benchmark: Futures+Copper					
h = 1	0.965**	0.969***	0.993	0.988*	0.971**	h = 1	0.96**	0.960***	1.002	0.987*	0.964***
h = 3	0.937***	0.942***	0.983	0.952**	0.924***	h = 3	0.939***	0.969***	1.001	0.991**	0.917***
h = 6	0.925***	0.934***	0.931**	0.949***	0.896***	h = 6	0.976***	1.001*	0.972*	0.994**	0.984*
h = 9	0.975***	0.984**	0.915**	1.008	0.911***	h = 9	1.031	1.045	0.963**	1.076	1.017
h = 12	0.963**	0.968***	0.952**	0.987**	0.907***	h = 12	0.994	1.009	0.965**	1.012	0.969*
Benchmark: Non-Oil+Copper						Benchmark: Non-Oil+Copper					
h = 1	0.969***	0.97***	0.997	0.987**	0.982***	h = 1	0.980***	0.965***	1.011	0.993*	0.976***
h = 3	0.944***	0.947***	1.000	0.958***	0.941***	h = 3	0.962***	0.973**	1.023	0.997	0.955***
h = 6	0.929***	0.934***	0.966***	0.949***	0.925***	h = 6	0.990	1.007	1.022	0.990**	1.015
h = 9	0.982**	0.997	0.957***	1.019	0.954**	h = 9	1.061	1.071	1.040	1.075	1.067
h = 12	0.987**	0.998	1.001	1.014	0.971**	h = 12	1.036	1.042	1.010	1.023	1.032

Notes: Ar(1), RW, DRW, Futures and Non-Oil represents the benchmarks models described in Table 1A and Table 1B. They are all augmented with a contemporaneous copper return. Inference about predictive ability is carried out with the Clark and West (2007) test, comparing forecasts generated with the models with and without Chilean Peso returns. "h" represents the forecast horizon in months. Figures lower than 1 favor the model with Chilean Peso. Figures higher than one favor the benchmarks without the Chilean Peso. P represents the number of one-step-ahead forecasts and R the sample size of the first estimation window in the out-of-sample exercise. Multistep-ahead–forecasts are computed using the direct method. **$p < 5\%$. ***$p < 1\%$.
Source: Authors' elaboration.

this subsection we contribute to answer this question by showing that the Chilean Peso has an interesting ability to predict WTI realized volatility and that this realized volatility is negatively correlated with fuels returns. This suggests that at least part of the predictive power shown by the Chilean peso flows through a volatility channel.

Table 15 shows results of the following regressions:

$$\Delta ln\left(FP_{t,t+h}\right) = a^{(h)} + \delta^{(h)}\left(RV_WTI_{t,t+h}\right) + u_{t+h} \tag{13}$$

for h = 1, 3, 6, 9 and 12 months. In (13) $RV_WTI_{t,\,t+h}$ denotes the realized volatility of WTI during the time period t, t + h. This volatility measure is computed as the sample standard deviation of daily WTI log returns during the time period t, t + h.

Table 15 shows that the parameters $\delta^{(h)}$ are statistically significant for all fuels and horizons. Their estimates are always negative, and the coefficients of determination vary from 2.9% to 23.2%. This relationship implies that if the Chilean Peso is successful at predicting $RV_WTI_{t,\,t+h}$ it could also be successful at predicting fuel returns via a volatility channel.

Table 16 shows estimates of the following specifications:

$$RV_WTI_{t,t+h} = \alpha^{(h)} + \beta^{(h)}\left[\Delta ln(ER_{t-1}) + \ldots + \Delta ln(ER_{t-3})\right]$$

$$+ \sum_{i=1}^{p} \varphi_i^{(h)} RV_WTI_{t-i,t+1-i} + u_{t+h} \tag{14}$$

where ER_t represents the Chilean nominal peso. Expression (14) is basically an augmented version of an Autoregressive Distributed Lag model of order p (ARDL(p)) in which realized volatility of a given fuel in the period t, t + h is regressed on p lags of one-month WTI realized volatility. The model is augmented with quarterly returns of the Chilean Peso. We consider two specifications, one in which p is set to 3 (top panel of Table 16), and other in which p is set to 12 (bottom panel of Table 16). In Table 16 we only report estimates of the coefficients associated to the Chilean Peso. We see that all of them are positive and statistically significant. The positive sign of the coefficients is consistent with all those negative coefficients reported in Section 4 and with the negative relationship between realized volatility and fuel returns. So a depreciation of the Chilean Peso predicts higher fuel volatility and hence lower returns. In other words, the volatility predictive channel is also consistent with the negative signs reported in the predictive relationship between Chilean exchange rate returns and fuels returns.

Table 17 presents results when evaluating the ability of the Chilean peso to predict WTI realized volatility out-of-sample. The ENCNEW test is able to reject the null of no predictability across most of our forecasting horizons, both choices of the p parameter and different partitions of our sample. This means that at the population level the MSPE of the model with the Chilean peso is lower than that of the model without the Chilean peso. RMSPE ratios are mostly lower than one, (with 3 exceptions) indicating that at the sample level the Chilean Peso is useful to increase the accuracy of the basic ARDL(p) models in most of our exercises. These results are consistent with those of Table 16 and provide evidence of a volatility channel in the predictive relationship between the Chilean peso and fuel returns.

Table 15
Estimates of a contemporaneous regression between WTI realized volatility and fuel returns.

	WTI	Brent	Propane	Gasoline	Heat oil
h = 1					
RV_WTI (1)	−0.030***	−0.029***	−0.024*	−0.026***	−0.027***
	(0.007)	(0.008)	(0.015)	(0.007)	(0.008)
Constant	0.070***	0.070***	0.053	0.061***	0.064***
	(0.014)	(0.018)	(0.034)	(0.017)	(0.018)
Obs	242	242	242	242	242
R-squared	0.102	0.079	0.029	0.040	0.083
h = 3					
RV_WTI (3)	−0.079***	−0.078**	−0.067**	−0.066**	−0.076***
	(0.029)	(0.032)	(0.026)	(0.031)	(0.025)
Constant	0.190***	0.190***	0.154***	0.162**	0.186***
	(0.059)	(0.065)	(0.056)	(0.064)	(0.053)
Obs	240	240	240	240	240
R-squared	0.154	0.138	0.068	0.076	0.157
h = 6					
RV_WTI (6)	−0.133***	−0.132**	−0.133***	−0.108**	−0.138***
	(0.047)	(0.053)	(0.045)	(0.051)	(0.044)
Constant	0.323***	0.324***	0.302***	0.268**	0.340***
	(0.104)	(0.116)	(0.108)	(0.109)	(0.104)
Obs	237	237	237	237	237
R-squared	0.187	0.169	0.125	0.109	0.221
h = 9					
RV_WTI (9)	−0.168***	−0.170***	−0.177***	−0.140**	−0.182***
	(0.056)	(0.063)	(0.058)	(0.056)	(0.060)
Constant	0.415***	0.425***	0.405***	0.353***	0.455***
	(0.130)	(0.147)	(0.144)	(0.129)	(0.112)
Obs	234	234	234	234	234
R-squared	0.199	0.189	0.132	0.141	0.232
h = 12					
RV_WTI (12)	−0.191***	−0.199***	−0.201***	−0.160***	−0.215***
	(0.061)	(0.069)	(0.07)	(0.061)	(0.065)
Constant	0.478***	0.504***	0.464***	0.410***	0.544***
	(0.148)	(0.171)	(0.175)	(0.146)	(0.166)
Obs	231	231	231	231	231
R-squared	0.189	0.185	0.115	0.143	0.223

Notes: Table 15 shows estimates of the coefficients in Eq. (13). RV_WTI(k) represents WTI realized volatility within k months. WTI, Brent, Propane, Gasoline and Heat Oil represents accumulated log returns of these fuels within h months. HAC standard errors are computed according to Newey and West (1987, 1994). $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

6. Extensions and robustness

6.1. Chile v/s other commodity-currencies

In this subsection, we extend our analysis to other commodity-currencies frequently considered in the literature: the Australian Dollar, the Canadian Dollar and the Norwegian Krone. Table 18 through Table 20 in the Companion Appendix report in-sample R^2 using the five econometrics specifications from Tables 1A and 1B, for h = 1, 3, 6, 9 and 12 months. Entries in bold signal the highest R^2 among the four exchange rates in each exercise. Some features are worth to be mentioned. First, for h = 1, 3, and 6, the specifications using the Chilean Peso display the highest in-sample fit, with no exceptions whatsoever (75 out of 75 comparisons). Second, for longer horizons (h = 9 and 12) the Chilean Peso exhibits the best in-sample fit in 74% of the exercises (37 out of 50). Finally, this in-sample fit superiority can be remarkable in magnitude: R^2 in models using the Chilean Peso are sometimes twice the R^2 of the specifications using other commodity-currencies.

To save space, we do not report the 400 regressions here, although they are available upon request. Nevertheless, as an illustration, Table 18 next exhibits our in-sample estimates of model (5) in Table 1A, comparing the four commodity-currencies for one-step-ahead forecasts. First, for the case of Australia, none of the coefficients associated with the exchange rate are statistically significant at usual significance levels. For the case of Canada, only 3 out of 10 coefficients are significant (for Brent, Gasoline and Heating Oil), while for Norway we reject the null hypothesis.

only once (for Propane). In sharp contrast, the coefficients associated with the Chilean Peso are all significant with no exceptions whatsoever (10 out of 10). Finally, Table 19 next summarizes the out-of-sample performance of each exchange rate. To save space, we report exclusively the Average RMSPE ratios for our five fuels, and our five specifications. We show results for P/R = 0.6 and P/R = 2. On average, the Chilean Peso exhibits the lowest RMSPE ratio, for both P/R = 0.6 and P/R = 2. These results are mainly driven by the remarkable performance of the Chilean Peso at short horizons (h = 1,3 and 6), in which the models with the Chilean currency provide the highest accuracy. At longer horizons of 9 and 12 months, the models with the Chilean Peso perform well, although not always they are the most accurate; (they are either the best or the second-best). While we do not report every out-of-sample exercise (they are available upon request), we would like to write a few words about the most relevant results. First, using the Chilean Peso, about 82% of RMSPE ratios are lower than one (206 out of 250); in sharp contrast, the percentage of RMSPE lower than one for Australia, Canada, and Norway is 53% (132 out of 250), 48% (119 out of 250), and 34% (85 out of 250) respectively. Second, regarding predictability at the population level, we use the CW test to determine if the models containing exchange rates outperform their respective benchmarks. In these analyses, models that use the Chilean Peso reject the null of equal population MSPE in about 83% of the exercises (207 out of 250). For

Table 16
Forecasting WTI realized volatility with and ARDL(p) model and the Chilean Peso.

	RV_WTI_1	*RV_WTI_3*	*RV_WTI_6*	*RV_WTI_9*	*RV_WTI_12*
$p = 3$					
Dlog(CLP(−1),0,3)	3.651*** (1.202)				
Dlog(CLP(−3),0,3)		4.133*** (1.279)			
Dlog(CLP(−6),0,3)			4.346*** (1.47)		
Dlog(CLP(−9),0,3)				3.637*** (1.298)	
Dlog(CLP(−12),0,3)					2.56** (1.254)
Obs	239	237	234	231	228
R-squared	0.501	0.507	0.402	0.290	0.200
$p = 12$					
Dlog(CLP(−1),0,3)	3.571*** (1.103)				
Dlog(CLP(−3),0,3)		4.35*** (1.21)			
Dlog(CLP(−6),0,3)			4.508*** (1.473)		
Dlog(CLP(−9),0,3)				3.759*** (1.349)	
Dlog(CLP(−12),0,3)					2.567* (1.421)
Obs	231	229	226	223	220
R-squared	0.516	0.511	0.403	0.294	0.214

Notes: *RV_WTI_k* denotes WTI realized volatility during a period of k months. Dlog(CLP(−k),0,3) represents the k order lag of quarterly returns of the Chilean Peso. HAC standard errors are computed according to Newey and West (1987, 1994). $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

specifications equipped with either the Australian, Canadian, or Norwegian exchange rate, the comparable null hypotheses are rejected in only 47% (117 out of 250), 36% (89 out of 250), and 27% (67 out of 250) of the exercises, respectively.

In summary, The Chilean peso fares extremely well in this competition with other commodity-currencies, especially at short horizons of one, three and six months.

6.2. Multiple unknown break points and asymmetries

We also carried out two additional exercises, one to explore potential asymmetries in the predictive relationship between the Chilean Peso and fuel returns, and another to explore the existence of potential unknown break points. As emphasized by Rossi (2006) and Giacomini and Rossi (2010), some predictive relationships in economic forecasting may be quite unstable. In particular, Pincheira and Hardy (2019) show that the predictive relationship between commodities and commodity-currencies may vary significantly across different sequential regimes. To explore potential instabilities in the predictive relationship between fuel prices and the Chilean Peso, we allow for the possibility of multiple unknown breaks in the regression parameters. As commented by Pincheira and Hardy (2019), this is a simple case of "time-varying" coefficients.

Details of these complementary exercises are described in full in subsections 3.2 and 3.3 in the Companion Appendix. Our findings clearly suggest asymmetries in the predictive relationship between the Chilean peso and fuel returns. In particular, a depreciation predicts stronger movements in fuel returns relative to an appreciation of the Chilean peso. In terms of breaks, we do find evidence of multiple regimes in the predictive relationship between the Chilean peso and fuel returns. Despite these instabilities, the null of no Granger causality is rejected in multiple regimes. In particular, at least one of the two coefficients associated to the Chilean Peso is statistically significant in most regimes:

Table 17
Out-of-Sample results when forecasting WTI realized volatility with the Chilean Peso.

h	ENCNEW		RMSPE ratios	
	P/R = 0.6	P/R = 2	P/R = 0.6	P/R = 2
p = 3				
1	3.910***	18.061***	0.993*	0.949*
3	3.857***	27.007***	0.996	0.920*
6	3.814***	16.033***	1.004	0.959*
9	1.116	9.967***	1.019	0.975
12	0.864	4.188**	1.003	0.990
p = 12				
1	5.308***	14.913***	0.973**	0.960*
3	7.051***	24.206***	0.958**	0.933*
6	7.050***	12.634***	0.967*	0.970
9	2.901**	9.920***	0.997	0.972
12	1.380*	5.640***	0.996	0.982

Notes: Table 17 shows out-of-sample results when forecasting WTI realized volatility with and without Chilean Peso returns in specification (14). In the first two columns of results, statistical significance is measured with the ENCNEW test. In the last two columns we report RMSPE ratios and statistical significance is measured using the Clark and West (2007) test. RMSPE ratios lower than 1 favor the model with the Chilean Peso. Figures higher than one favor the benchmarks without the Chilean Peso. P represents the number of one-step-ahead forecasts and R the sample size of the first estimation window in the out-of-sample exercise. Multistep-ahead–forecasts are computed using the direct method. $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

5 out of 5 for WTI, 1 out of 1 for Propane, 4 out of 6 for Brent, 5 out of 6 for Gasoline, and 4 out of 5 for Heating Oil. Notably, for the case of WTI, both coefficients associated with the exchange rate are statistically significant in every regime. This is important since it indicates that our results are not driven only by the Global Financial Crisis or more generally by "small pockets of predictability".

7. Summary and concluding remarks

In the last decade a vast literature has evaluated the ability of commodity-currencies to predict commodity returns. In this paper we make a contribution to this literature by showing that the Chilean exchange rate has the ability to predict the returns of oil prices and of three additional oil-related products: gasoline, propane and heating oil. We show this with both in-sample and out-of sample exercises at multiple horizons, building forecasts one, three, six, nine and twelve months ahead.

Two natural explanations of our findings rely on the well know "dollar effect" and on the present-value theory for exchange rate determination in combination with the strong co-movement displayed by fuels and commodity prices. This second explanation is based on the fact that the Chilean economy is heavily influenced by one particular commodity: copper, which represents nearly 50% of total national exports and attracts a similar share in terms of Foreign Direct Investment. As a consequence, the floating Chilean exchange rate is importantly affected by fluctuations in the copper price. As oil-related products display an important co-movement with base metal prices, it is reasonable to expect evidence of Granger causality from the Chilean peso to these oil-related products.

Interestingly, in this paper we show that the predictive ability of the Chilean peso cannot only be justified on the grounds of the two natural explanations mentioned before. This is so, because our exercises with a multilateral exchange rate provide results with small differences relative to our exercises with the nominal Chilean Peso. We arrive to a similar conclusion regarding the importance of the Commodity Currency Hypothesis in our results. This is so, because we show that the predictive ability of the Chilean peso is only mildly overshadowed by the presence of a contemporaneous copper return in our specifications. If the most

Table 18
Forecasting fuel returns one-step-ahead with commodity-currencies.

In-sample analysis with the Non-Oil benchmark	WTI	Brent	Propane	Gasoline	Heating oil
Australia					
ER(−1) + ER(−2) + ER(−3)	−0.126	−0.176	−0.144	−0.304	−0.116
	(0.155)	(0.21)	(0.242)	(0.215)	(0.161)
ER(−4) + ER(−5) + ER(−6)	−0.090	−0.092	−0.088	−0.017	−0.145
	(0.118)	(0.121)	(0.137)	(0.129)	(0.106)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.154***	0.13**	0.113	0.067	0.146**
	(0.055)	(0.054)	(0.084)	(0.077)	(0.058)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.106**	−0.115**	−0.031	−0.100*	−0.086
	(0.05)	(0.045)	(0.067)	(0.056)	(0.053)
Constant	0.003	0.004	−0.002	0.004	0.003
	(0.005)	(0.005)	(0.007)	(0.004)	(0.005)
R-squared	0.074	0.060	0.038	0.042	0.069
Canada					
ER(−1) + ER(−2) + ER(−3)	0.118	0.114	0.158	0.258**	0.076
	(0.146)	(0.137)	(0.210)	(0.121)	(0.140)
ER(−4) + ER(−5) + ER(−6)	−0.171	−0.204*	−0.223	−0.046	−0.279*
	(0.159)	(0.122)	(0.183)	(0.146)	(0.162)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.232***	0.225***	0.206***	0.24***	0.212***
	(0.076)	(0.068)	(0.078)	(0.069)	(0.066)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.119*	−0.136**	−0.057	−0.117	−0.103*
	(0.063)	(0.063)	(0.065)	(0.076)	(0.057)
Constant	0.003	0.003	−0.002	0.003	0.003
	(0.005)	(0.005)	(0.007)	(0.005)	(0.005)
R-squared	0.075	0.059	0.026	0.035	0.074
Norway					
ER(−1) + ER(−2) + ER(−3)	−0.214	−0.218	−0.331*	−0.177	−0.191
	(0.151)	(0.171)	(0.186)	(0.176)	(0.125)
ER(−4) + ER(−5) + ER(−6)	−0.016	−0.021	−0.112	0.079	−0.132
	(0.091)	(0.088)	(0.168)	(0.079)	(0.083)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.144**	0.137***	0.072	0.131***	0.138***
	(0.057)	(0.046)	(0.062)	(0.048)	(0.048)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.081*	−0.092**	−0.033	−0.079*	−0.074*
	(0.047)	(0.043)	(0.069)	(0.048)	(0.043)
Constant	0.003	0.004	0.000	0.003	0.004
	(0.004)	(0.004)	(0.006)	(0.004)	(0.005)
R-squared	0.080	0.062	0.033	0.035	0.074
Chile					
ER(−1) + ER(−2) + ER(−3)	−0.303***	−0.304**	−0.306**	−0.336***	−0.233**
	(0.112)	(0.119)	(0.130)	(0.126)	(0.113)
ER(−4) + ER(−5) + ER(−6)	−0.516***	−0.614***	−0.422*	−0.582**	−0.474***
	(0.15)	(0.211)	(0.234)	(0.273)	(0.164)
LMEX(−1) + LMEX(−2) + LMEX(−3)	0.057	0.043	0.030	0.013	0.076
	(0.045)	(0.05)	(0.064)	(0.052)	(0.047)
LMEX(−4) + LMEX(−5) + LMEX(−6)	−0.193***	−0.23***	−0.091	−0.232**	−0.149***
	(0.063)	(0.08)	(0.072)	(0.107)	(0.056)
Constant	0.008**	0.01**	0.003	0.009***	0.008*
	(0.004)	(0.004)	(0.006)	(0.003)	(0.004)
R-squared	0.141	0.128	0.044	0.078	0.119

Notes: WTI, Brent, Propane, Gasoline and Heating Oil represent one-month returns of each fuel. LMEX denotes London Metal Exchange index returns. HAC standard errors are computed according to Newey and West (1987, 1994). $^{**}p < 5\%$. $^{***}p < 1\%$.
Source: Authors' elaboration.

Table 19
Average RMSPE ratios across fuels and models.

	Australia	Canada	Norway	Chile
P/R = 0.6				
h = 1	0.996	1.000	0.998	0.976
h = 3	0.99	1.000	0.998	0.945
h = 6	0.974	0.953	0.989	0.922
h = 9	0.984	0.952	1.035	0.981
h = 12	0.974	0.990	1.035	0.986
Average	0.983	0.979	1.011	0.962
P/R = 2				
h = 1	0.996	1.006	0.986	0.958
h = 3	1.002	1.016	0.998	0.943
h = 6	1.040	1.019	1.009	0.94
h = 9	1.074	1.001	1.031	0.988
h = 12	1.085	1.008	1.043	1.016
Average	1.039	1.01	1.013	0.969

Source: Authors' elaboration.

important predictive channel were copper prices, then the inclusion of a contemporaneous copper term should kill or at least reduce importantly the predictive ability of the Chilean Peso.

We also show another plausible predictive channel: volatility. We report an important contemporaneous negative correlation between fuels returns and WTI realized volatility altogether with an interesting ability of Chilean Peso to predict this WTI realized volatility. Thus, the combination of both effects is a new plausible predictive channel from the Chilean peso to fuel returns.

We also carry out predictive exercises with other commodity-currencies: the Australian Dollar, the Canadian Dollar and the Norwegian Krone. While these currencies do have the ability to predict fuel returns, our results indicate that the predictability stemming from the Chilean Peso is stronger, especially at short horizons.

Our results are consistent with the studies of Alquist et al. (2013), Gillman and Nakov (2009), Funk (2018), Baumeister and Kilian (2015)

and Garratt et al. (2019) that have also shown that crude oil returns are indeed predictable. They are also consistent with the results in Alquist et al. (2013) that show predictive ability from the Australian and Canadian dollars to crude oil returns. Our results also resemble those in Pincheira and Hardy (2019) in two dimensions: first by showing predictability from a commodity-currency to returns of commodities that are not key elements in the export basket of the relevant country, and second, by reporting strong evidence of predictability at the population level, yet moderate evidence at the sample level.

Our results suggests that the strong predictive ability of the Chilean Peso has to do with three different channels that work in the same direction: the traditional "dollar effect", the CCH in combination with comovement between copper and fuel returns and the volatility channel in combination with a contemporaneous leverage effect in fuel returns. These three channels generate an inverse predictive relationship between the Chilean Peso and fuel returns, which is consistent with our in-sample estimates in Section 4.

We have reported two results that, to our knowledge, are new in the literature: a) the volatility channel, and b) the ability of the Chilean Peso to predict fuel returns beyond the implications of the CCH. In the case of the Chilean currency this last exercise is easy to implement, given the strong dependence of the Chilean economy with only one single commodity: copper. We leave as an extension for future research the implementation of these two new exercises in the context of different combinations of commodities and commodity-currencies.

Declaration of Competing Interest

We have no competing Interests.

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Appendix A. Appendix

Table A1
Correlations in logarithmic price levels.

	Copper	WTI	WTI futures	Brent	Propane	Gasoline	Heating oil
Copper	1.00	0.91	0.92	0.92	0.69	0.93	0.93
WTI	0.91	1.00	1.00	0.99	0.83	0.97	0.98
WTI Futures	0.92	1.00	1.00	0.99	0.83	0.97	0.98
Brent	0.92	0.99	0.99	1.00	0.79	0.98	0.99
Propane	0.69	0.83	0.83	0.79	1.00	0.77	0.80
Gasoline	0.93	0.97	0.97	0.98	0.77	1.00	0.98
Heating oil	0.93	0.98	0.98	0.99	0.80	0.98	1.00

Source: Authors' elaboration.

Table A2
Correlations in logarithmic differences.

	Copper	WTI	WTI futures	Brent	Propane	Gasoline	Heating oil
Copper	1.00	0.44	0.45	0.41	0.29	0.37	0.40
WTI	0.44	1.00	0.99	0.87	0.50	0.70	0.81
WTI futures	0.45	0.99	1.00	0.88	0.51	0.71	0.82
Brent	0.41	0.87	0.88	1.00	0.49	0.72	0.79
Propane	0.29	0.50	0.51	0.49	1.00	0.42	0.69
Gasoline	0.37	0.70	0.71	0.72	0.42	1.00	0.66
Heating oil	0.40	0.81	0.82	0.79	0.69	0.66	1.00

Source: Authors' elaboration.

Table A3
Descriptive statistics of the monthly returns of the series.

	WTI	WTI futures	Brent	Propane	Gasoline	Heating oil	LME index	CLP
Mean	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00
Median	0.02	0.02	0.01	0.00	0.01	0.01	0.00	0.00
Std. Dev.	0.09	0.09	0.11	0.14	0.13	0.09	0.06	0.03
Max	0.28	0.25	0.33	0.58	0.67	0.24	0.20	0.20
Min	−0.39	−0.38	−0.44	−0.90	−0.47	−0.42	−0.33	−0.07
Obs	242	242	242	242	242	242	242	242

Source: Authors' elaboration.

Table A4
Descriptive statistics of the monthly returns of the series.

	ECLP	AUD	CAD	NOK	Copper
Mean	0.00	0.00	0.00	0.00	0.01
Median	0.00	0.00	0.00	0.00	0.01
Std. Dev.	0.03	0.04	0.03	0.03	0.07
Max	0.13	0.18	0.13	0.13	0.27
Min	−0.09	−0.09	−0.08	−0.07	−0.47
Obs	242	242	242	242	242

Source: Authors' elaboration.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2021.105802.

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